# CSE 708
# Graph Theory

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The Prufer Sequence is one of the proofs of cayley's theorem. It relies on prufers sequence. This proof establishes a flawless bijection between every possible labeled tree of n vertices.

So we will turn a labeled tree into sequence, and follow a simple algorithmic rule repeatedly untill only two vertices left.

1) Find the leaf (vertex connected to one edge)
2) look at its neighbors and write their sequence
3) Remove the leaf and edge from the tree
4) Repeat this process untill only two vertices remain.

Because we stsrt with n vertices and stop when 2 vertices are left, exactly 2(n-2) vertices are deleted. This means when writing down sequence only n-2 numbers will remain.